

Adaptive AI Monitoring Platform

Drift Analysis Report — Job ID: 4be251a1-d53a-444f-b7b6-bb3bbafc8677

Reference Dataset: v1 (ID: dfa60168)

Target Dataset: v1 (ID: 3036d0a2)

Date Run: 2026-07-08 02:39:42

Status: COMPLETED

AI-Generated Analysis Summary

A comparison between Reference Version (v1) and Target Version (v1) revealed that 6 features show statistically significant drift. Particularly, critical features like Time, V12, V14, Amount exhibit major drift severity, indicating a potential decrease in model performance.

Key Findings:

- Business Impact: The detected distribution shift may reduce model precision/recall stability in production if left untreated. Risk increases when major-severity features are drifted.
- Feature 'Time' exhibits major data drift (PSI/Chi-Sq Score: 0.9816, p-value: 0.0000).
- Feature 'V11' exhibits none data drift (PSI/Chi-Sq Score: 0.0247, p-value: 0.0097).
- Feature 'V12' exhibits major data drift (PSI/Chi-Sq Score: 1.2957, p-value: 0.0000).
- Feature 'V14' exhibits major data drift (PSI/Chi-Sq Score: 0.7040, p-value: 0.0000).
- Feature 'V18' exhibits none data drift (PSI/Chi-Sq Score: 0.0305, p-value: 0.0062).
- Feature 'Amount' exhibits major data drift (PSI/Chi-Sq Score: 0.4111, p-value: 0.0000).

Actionable Recommendations:

- Retrain the model using target version dataset to adjust for shifts in Time, V12, V14, Amount.
- Investigate upstream data collection pipelines to verify if data schema or collection mechanisms changed.

Feature Drift Details

Feature Name	Statistical Method	PSI Score	p-value	Drift Status	Severity
Time	Kolmogorov-Smirnov & PSI	0.9816	0.0000	DRIFTED	MAJOR
V1	Kolmogorov-Smirnov & PSI	0.0176	0.5363	Stable	NONE
V2	Kolmogorov-Smirnov & PSI	0.0194	0.1812	Stable	NONE
V3	Kolmogorov-Smirnov & PSI	0.0157	0.1812	Stable	NONE
V4	Kolmogorov-Smirnov & PSI	0.0130	0.4006	Stable	NONE
V5	Kolmogorov-Smirnov & PSI	0.0315	0.6478	Stable	NONE
V6	Kolmogorov-Smirnov & PSI	0.0078	0.8882	Stable	NONE
V7	Kolmogorov-Smirnov & PSI	0.0148	0.2407	Stable	NONE
V8	Kolmogorov-Smirnov & PSI	0.0297	0.2194	Stable	NONE
V9	Kolmogorov-Smirnov & PSI	0.0221	0.1996	Stable	NONE
V10	Kolmogorov-Smirnov & PSI	0.0147	0.8282	Stable	NONE
V11	Kolmogorov-Smirnov & PSI	0.0247	0.0097	DRIFTED	NONE
V12	Kolmogorov-Smirnov & PSI	1.2957	0.0000	DRIFTED	MAJOR
V13	Kolmogorov-Smirnov & PSI	0.0209	0.2407	Stable	NONE
V14	Kolmogorov-Smirnov & PSI	0.7040	0.0000	DRIFTED	MAJOR
V15	Kolmogorov-Smirnov & PSI	0.0140	0.6102	Stable	NONE
V16	Kolmogorov-Smirnov & PSI	0.0151	0.7228	Stable	NONE
V17	Kolmogorov-Smirnov & PSI	0.0069	0.8282	Stable	NONE
V18	Kolmogorov-Smirnov & PSI	0.0305	0.0062	DRIFTED	NONE
V19	Kolmogorov-Smirnov & PSI	0.0242	0.4326	Stable	NONE

V20	Kolmogorov-Smirnov & PSI	0.0037	0.8282	Stable	NONE
V21	Kolmogorov-Smirnov & PSI	0.0199	0.2878	Stable	NONE
V22	Kolmogorov-Smirnov & PSI	0.0276	0.7228	Stable	NONE
V23	Kolmogorov-Smirnov & PSI	0.0210	0.0869	Stable	NONE
V24	Kolmogorov-Smirnov & PSI	0.0215	0.2407	Stable	NONE
V25	Kolmogorov-Smirnov & PSI	0.0192	0.6855	Stable	NONE
V26	Kolmogorov-Smirnov & PSI	0.0231	0.6855	Stable	NONE
V27	Kolmogorov-Smirnov & PSI	0.0186	0.7228	Stable	NONE
V28	Kolmogorov-Smirnov & PSI	0.0173	0.9137	Stable	NONE
Amount	Kolmogorov-Smirnov & PSI	0.4111	0.0000	DRIFTED	MAJOR
Class	Kolmogorov-Smirnov & PSI	0.0000	1.0000	Stable	NONE

AI Safety & Factuality Guardrails Log

Check: input_safety | **Status:** PASS | **Score:** 1.00

Audit Feedback: Input safety guardrail passed.

Check: output_conformity_and_factuality | **Status:** PASS | **Score:** 1.00

Audit Feedback: All output schema and statistical checks passed.